

MEHRAB JALILIAN



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ABOUT ME

Data Scientist with hands-on experience developing intelligent, scalable AI solutions in climate control, healthcare, and NLP. Skilled at deploying ML systems that improve performance, optimize outcomes, and deliver measurable results. Ethical hacking and bug bounty.

EDUCATION

Bachelor of Computer Engineering

Islamic Azad University

- Coursework: Machine Learning, Deep Learning, Data Structures, Statistics
- Final Thesis: "Fuzzy Logic-Based Climate Control in Smart Greenhouses"
- Member of Data Science Research Group & AI Lab
- Research Director, Computer Science Scientific Association

EXPERIENCE

- Anomaly Detection for Home Mining Devices: Developed an intelligent anomaly detection system to identify unauthorized home cryptocurrency miners in residential networks. Leveraged machine learning algorithms (e.g., isolation forests and autoencoders) on network traffic data to detect irregular patterns, achieving high precision in flagging suspicious devices while minimizing false positives. This scalable solution enhanced network security and resource optimization for home users.
- Anomaly Detection for Electrical Anomalies in Industry: Designed a robust ML-based anomaly detection framework for monitoring electrical irregularities in industrial power systems. Analyzed time-series data from sensors to identify faults like voltage spikes or equipment failures using techniques such as LSTM networks and statistical modeling. Deployed the system to improve predictive maintenance, reducing downtime by detecting anomalies in real-time and delivering actionable insights for operational efficiency.
- Improving Growth Functions in Language Models: Optimized growth functions in large language models to enhance training efficiency and performance scaling. Explored modifications to activation functions and parameter growth strategies (e.g., inspired by scaling laws) using frameworks like PyTorch, resulting in faster convergence and improved model accuracy on NLP tasks. This work contributed to more resource-efficient AI development, with measurable gains in inference speed and output quality.
- Predictive Maintenance for Renewable Energy Systems: Built a predictive analytics model using time-series forecasting and anomaly detection to anticipate failures in solar panel arrays. Integrated LSTM and Transformer-based architectures on sensor data from IoT devices, achieving 25% reduction in unplanned downtime and 15% improvement in energy efficiency. Deployed on AWS with real-time monitoring, contributing to sustainable energy optimization for utility-scale installations.
- Bug Bounty & Web Application Security Research: Participated in bug bounty programs on platforms such as HackerOne and Bugcrowd, successfully identifying and responsibly disclosing security vulnerabilities in live web applications. Focused on common issues including XSS, IDOR, and authentication flaws using manual testing and tools like Burp Suite. Gained practical experience in vulnerability assessment, ethical hacking, and secure reporting, enhancing system security for target organizations while developing strong analytical and problem-solving skills transferable to data security tasks.

SKILLS

Languages & Tools Python • SQL • Bash • Git • Jupyter

Libraries & Frameworks NumPy • Pandas • Scikit-learn • TensorFlow • Keras • MLflow

AI & ML Expertise Supervised/Unsupervised Learning • Neural Networks • SHAP • Evaluation Metrics

Specialized Domains Computer Vision • NLP • Time Series • Climate Forecasting • Healthcare AI

Deployment & Visualization FastAPI • Docker • Matplotlib

Workflow & Collaboration Agile • Scrum • GitHub • Research Coordination

Web Application Security Testing, Vulnerability Assessment & Penetration Testing, Ethical Hacking and Responsible Disclosure

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Bug Bounty Hunting (HackerOne, Bugcrowd, Intigriti), OWASP Top 10, Secure Reporting & Analytical Problem-Solving